AMANUR YOOSUF

Ho Man Tin, Kowloon, Hong Kong | +852 9763 0974 | amanur.yoosuf@connect.polyu.hk | amanyoosuf.com

SUMMARY

Electrical Engineering and AI & Data Analytics undergraduate at PolyU who builds AI-driven software end to end: LSTM forecasting models in Python and React web interfaces. Conducting URIS-funded research on deep learning dynamic state estimation for power grids, running a personal portfolio website (amanyoosuf.com) hosted on GitHub, and completing Harvard's CS50P. Keen to integrate LLMs into practical products and bridge AI research with real deployment in a collaborative engineering team.

EDUCATION

The Hong Kong Polytechnic University, Hung Hom, Hong Kong **June 2029 (Expected)**
B.Eng (Hons.) in Electrical Engineering, with AI and Data Analytics (AIDA)
CGPA: 3.53/4.30

- Awarded Undergraduate Research and Innovation Scheme (URIS) Scholarship worth HK$50,000, a competitive university-wide programme promoting undergraduate research
- Awarded Full Entry Scholarship worth HK$900,000 (merit-based, awarded to top non-local entrants)
- Awarded Talent Development Scholarship (25/26) worth HK$10,000 for involvement in extracurricular activities

Lyceum International School, Colombo, Sri Lanka **June 2024**
Edexcel International Advanced Levels (IAL): Biology, Chemistry, Physics, Mathematics.
Awards: Cambridge Outstanding Learner Award; Edexcel High Achiever Award

UNDERGRADUATE RESEARCH

URIS Research Project, Dept. of Electrical and Electronic Engineering, PolyU **Sep 2026 – Present**
Title: Data-Driven Dynamic State Estimation for Situational Awareness of Modern Power Grids
Supervisor: Prof. Siqi Bu

- Developing a deep learning (LSTM) model in PyTorch/TensorFlow to forecast bus voltages from sparse measurements in distribution grids with high EV and solar penetration
- Building a Python/Pandapower simulation framework on the IEEE 33-bus feeder to generate ground-truth data; benchmarking against persistence, static DNN and WLS baselines
- Evaluating robustness under measurement loss and bad-data injection; designing a visual alert module that turns forecasts into colour-coded early warnings for operators, bridging AI research and practical use

PROJECTS

Personal Portfolio Website (amanyoosuf.com) **2026 – Present**

- Designed and built a responsive site in React, JavaScript, HTML and CSS with reusable components, scroll animations and dark/light mode; version-controlled with Git and hosted on GitHub

CS50P: Harvard's Introduction to Programming with Python **In progress**

- Completing problem sets and projects in Python, with code committed and managed on GitHub

Google Maps API Data Pipeline, Research Assistantship, PolyU CEE Department **Jan 2026 – March 2026**

- Managed data collection across 1,000+ API queries with error-free retrieval and validation; documented processing steps and quality checks

Circuit Calculator **Dec 2025**

- Built a Python circuit calculator using NumPy & Matplotlib to automate calculations and visualise outputs; verified across 10+ circuit types through systematic debugging and testing

TECHNICAL/PART-TIME EXPERIENCE

STEM Project Developer, Scholastic Academy, Central, Hong Kong (Part-time) **Sep 2025 – Present**

- Prototyped interactive digital learning tools and proof-of-concept solutions using AI tools and automation workflows for students aged 6–12
- Analysed usage data to drive R&D and inform iterative improvements to internal systems and learner experience

PolyU E-Formula EE Sub-Team Member, Hung Hom, Hong Kong (Part-time) **Oct 2025 – June 2026**

- Supported design, testing and optimisation of electrical systems including wiring, sensors and motor control
- Conducted functional testing and debugging of embedded electronics, documenting test results and producing progress reports

SKILLS

- Programming & Web: Python (NumPy, Pandas, SciPy, Matplotlib), JavaScript, React, HTML/CSS, Git/GitHub, MATLAB, Simulink, VS Code
- AI & Machine Learning: PyTorch/TensorFlow, DNN/LSTM, time-series forecasting, Pandapower, Jupyter Notebook; AI tools (Claude, Copilot, Gemini)
- Hardware: Embedded electronics testing, sensors, motor control
- Languages: English (Native)